For the use only of a Registered Medical Practitioner

PRESCRIBING INFORMATION

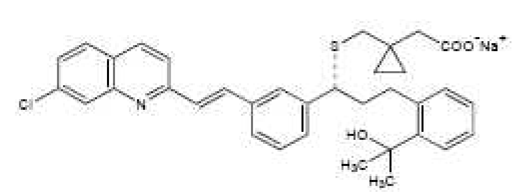
AIRLUKAST TABLETS
(Montelukast Sodium Tablets 10 mg)

COMPOSITION
Each film coated tablet contains:
Montelukast Sodium equivalent to
Montelukast ………………………10 mg

PRODUCT DESCRIPTION
Beige to light yellow coloured, rounded square, film-coated tablets, engraved with '10' on one side and scored on the other side.

DESCRIPTION

AIRLUKAST contain montelukast sodium, which is a selective and orally active leukotriene receptor antagonist that inhibits the cysteinyl leukotriene $CysLT_1$ receptor. It is chemically described as [R-(E)]-1-[[[1-[3-[2-(7-chloro-2-quinolinyl) ethenyl] phenyl]-3-[2-(1-hydroxy-1-methyl ethyl)phenyl]propyl]thio]methyl]cyclopropane acetic acid, monosodium salt. Its empirical formula is $C_{35}H_{35}ClNNaO_3S$, and its molecular weight is 608.18. The structural formula is:



MONTELUKAST SODIUM

PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES

Pharmacodynamics
The cysteinyl leukotrienes (LTC4, LTD4, LTE4) are products of arachidonic acid metabolism and are released from various cells, including mast cells and eosinophils. These eicosanoids bind to cysteinyl leukotriene (CysLT) receptors. The CysLT type-1 (CysLT1) receptor is found in the human airway (including airway smooth muscle cells and airway macrophages) and on other pro-inflammatory cells

(including eosinophils and certain myeloid stem cells). CysL Ts have been correlated with the pathophysiology of asthma and allergic rhinitis. In asthma, leukotriene-mediated effects include airway edema, smooth muscle contraction, and altered cellular activity associated with the inflammatory process. In allergic rhinitis, CysL Ts are released from the nasal mucosa after allergen exposure during both early-and late-phase reactions and are associated with symptoms of allergic rhinitis.

Montelukast is an orally active compound that binds with high affinity and selectivity to the CysLT1 receptor (in preference to other pharmacologically important airway receptors, such as the prostanoid, cholinergic, or beta-adrenergic receptor). Montelukast inhibits physiologic actions of LTD4 at the CysLT1 receptor without any agonist activity.

Montelukast causes inhibition of airway cysteinyl leukotriene receptors as demonstrated by the ability to inhibit bronchoconstriction due to inhaled LTD4 in asthmatics. Doses as low as 5 mg cause substantial blockage of LTD4-induced bronchoconstriction.

Pharmacokinetics

Montelukast is rapidly absorbed following oral administration. For the 10 mg film-coated tablet, the mean peak plasma concentration (C_{max}) is achieved 3 hours (T_{max}) after administration in adults in the fasted state. The mean oral bioavailability is 64%. The oral bioavailability and C_{max} are not influenced by a standard meal.

Montelukast is more than 99% bound to plasma proteins. The steady-state volume of distribution of montelukast averages 8-11 litres. Minimal distribution of radiolabelled montelukast across the blood-brain barrier has been reported in rats. In addition, concentrations of radiolabelled material at 24 hours post-dose were minimal in all other tissues.

Montelukast is extensively metabolised. With therapeutic doses, plasma concentrations of metabolites of montelukast are undetectable at steady state in adults and children.

In reported *in vitro* studies using human liver microsomes indicate that cytochrome P450 3A4, 2A6 and 2C9 are involved in the metabolism of montelukast. However, therapeutic plasma concentrations of montelukast do not inhibit cytochromes P450 3A4, 2C9, 1A2, 2A6, 2C19, or 2D6 in human liver microsomes. The contribution of metabolites to the therapeutic effect of montelukast is minimal.

The plasma clearance of montelukast averages 45 ml/min in healthy adults. Following an oral dose of radiolabelled montelukast, 86% of the radioactivity was recovered in 5-day faecal collections and <0.2% was recovered in urine. Coupled with estimates of montelukast oral bioavailability, this indicates that montelukast and its metabolites are excreted almost exclusively via the bile.

The mean plasma half-life of montelukast ranged from 2.7 to 5.5 hours in healthy young adults. The pharmacokinetics of montelukast are nearly linear for oral doses up to 50 mg. During once-daily dosing with 10-mg montelukast, there is little accumulation of the parent drug in plasma (14%).

Pharmacokinetics in special populations

Gender: The pharmacokinetics of montelukast is similar in males and females.

Race: Pharmacokinetic differences due to race have not been studied.

Hepatic Insufficiency: Patients with mild-to-moderate hepatic insufficiency and clinical evidence of cirrhosis had evidence of decreased metabolism of montelukast resulting in 41% (90% CI=7%, 85%) higher mean montelukast area under the plasma concentration curve (AUC) following a single 10-mg dose. The elimination of montelukast was slightly prolonged compared with that in healthy subjects (mean half-life, 7.4 hours). No dosage adjustment is required in patients with mild-to-moderate hepatic insufficiency. The pharmacokinetics of montelukast in patients with more severe hepatic impairment or with hepatitis have not been evaluated.

Renal Insufficiency: Since montelukast and its metabolites are not excreted in the urine, the pharmacokinetics of montelukast were not evaluated in patients with renal insufficiency. No dosage adjustment is recommended in these patients.

Adolescents: The plasma concentration profile of montelukast following administration of the 10-mg film-coated tablet is similar in adolescents ≥15 years of age and young adults.

INDICATIONS / USAGE

- Prophylaxis and chronic treatment of asthma in adults.
- For the relief of day-time and night-time symptoms of seasonal allergic rhinitis in adults.

DOSE AND METHOD OF ADMINISTRATION

Montelukast should be taken once daily in the evening. For asthma, montelukast should be taken in evening. For allergic rhinitis, the time of administration may be individualized to suit patient needs.

Patients with both asthma and allergic rhinitis should take only one tablet daily in the evening.

The dosage for asthma and/or seasonal allergic rhinitis

- Adults and adolescents 15 years of age and older is 10-mg once daily.

The therapeutic effect of montelukast on parameters of asthma control occurs within one day. Montelukast may be taken with or without food. Patients should be advised to continue taking montelukast even if their asthma is under control, as well as during periods of worsening asthma.

No dosage adjustment is necessary for the elderly, or for patients with renal insufficiency, or mild to moderate hepatic impairment. The dosage is the same for both male and female patients. There is no information on patients with severe hepatic impairment.

Therapy with montelukast in relation to other treatments for asthma.
Montelukast can be added to a patient's existing treatment regimen.

Inhaled corticosteroids: Treatment with montelukast can be used as add-on therapy in patients when inhaled corticosteroids plus "as needed" short acting β-agonists provides inadequate clinical control. Montelukast should not be abruptly substituted for inhaled corticosteroids.

CONTRAINDICATIONS

Hypersensitivity to montelukast or to any of the excipients of the product

WARNINGS AND PRECAUTIONS

Montelukast is not indicated for use in the reversal of bronchospasm in acute asthma attacks, including status asthmaticus. Patients should be advised to have appropriate rescue medication available. Therapy with montelukast can be continued during acute exacerbations of asthma. Patients who have exacerbations of asthma after exercise should have available for rescue a short-acting inhaled beta-agonist.

While the dose of inhaled corticosteroid may be reduced gradually under medical supervision, montelukast should not be abruptly substituted for inhaled or oral corticosteroids.

Patients with known aspirin sensitivity should continue avoidance of aspirin or non-steroidal anti-inflammatory agents while taking montelukast. Although montelukast is effective in improving airway function in asthmatics with documented aspirin sensitivity, it has not been shown to truncate bronchoconstrictor response to aspirin and other non-steroidal anti-inflammatory drugs in aspirin-sensitive asthmatic patients.

Neuropsychiatric events have been reported in adult, adolescent, and paediatric patients taking montelukast. Agitation, aggressive behaviour or hostility, anxiousness, depression, dream abnormalities, hallucinations, insomnia, irritability, restlessness, somnambulism, suicidal thinking and behaviour (including suicide), and tremor have been reported with the use of montelukast. The clinical details of some these adverse events involving montelukast appear consistent with a drug-induced effect.

Patients and prescribers should be alert for neuropsychiatric events. Patients should be instructed to notify their prescriber if these changes occur. Prescribers should carefully evaluate the risks and benefits of continuing treatment with montelukast if such events occur.

Patients with asthma on therapy with montelukast may present with systemic eosinophila, sometimes presenting with clinical features of vasculitis consistent with Churg-Strauss syndrome, a condition which is often treated with systemic corticosteroid therapy. These events usually, but not always, have been associated with the reduction of oral corticosteroid therapy. Physicians should be alert to eosinophilia, vasculitic rash, worsening pulmonary symptoms, cardiac complications, and/or neuropathy presenting in their patients. A causal association between montelukast and these underlying conditions has not been established.

AIRLUKAST CHEWABLE TABLETS 10 mg contains lactose. Patients with rare hereditary problems of galactose intolerance, total lactase deficiency or glucosegalactose malabsorption should not take this medicine.

DRUG INTERACTIONS

Montelukast may be administered with other therapies routinely used in the prophylaxis and chronic treatment of asthma. It has been reported that the recommended clinical dose of montelukast did not have clinically important effects on the pharmacokinetics of the following medicinal products: theophylline, prednisone, prednisolone, oral contraceptives (ethinyl estradiol/ norethindrone 35/1), terfenadine, digoxin and warfarin.

The area under the plasma concentration curve (AUC) for montelukast was decreased approximately 40% in subjects with co-administration of phenobarbital. Since montelukast is metabolised by CYP 3A4, caution should be exercised, particularly in children, when montelukast is co-administered with inducers of CYP 3A4, such as phenytoin, phenobarbital and rifampicin.

Montelukast is a potent inhibitor of CYP 2C8 *in vitro*. However, montelukast does not inhibit CYP 2C8 *in vivo*. Therefore, montelukast is not anticipated to markedly alter the metabolism of medicinal products metabolised by this enzyme (e.g., paclitaxel, rosiglitazone, and repaglinide).

SIDE EFFECTS /ADVERSE REACTIONS

Montelukast has been generally well tolerated. Side effects of which were mild, generally did not required discontinuation of therapy.

The following drug-related adverse reactions have been reported commonly in asthmatic patients with montelukast

Body System Class	Adult Patients 15 years and older
Nervous system disorders	headache
Gastro-intestinal disorders	abdominal pain

The following adverse reactions have been reported in patients with montelukast during post marketing use:

Infections and infestations: upper respiratory infection.

Postmarketing Experience
Blood and lymphatic system disorders: thrombocytopenia, increased bleeding tendency.

Immune system disorders: hypersensitivity reactions including anaphylaxis, hepatic eosinophilic infiltration.

Psychiatric disorders: dream abnormalities including nightmares, hallucinations, insomnia, somnambulism, irritability, anxiety, restlessness, agitation including aggressive behavior or hostility, tremor, depression, obsessive-compulsive symptoms, suicidal thinking and behavior (suicidality) in very rare cases.

Nervous system disorders: dizziness drowsiness, paraesthesia/hypoesthesia, seizure.

Cardiac disorders: palpitations.

Respiratory, thoracic and mediastinal disorders: epistaxis

Gastro-intestinal disorders: diarrhea, dry mouth, dyspepsia, nausea, vomiting.

Hepatobiliary disorders: elevated levels of serum transaminases (ALT, AST), hepatitis (including cholestatic, hepatocellular, and mixed-pattern liver injury).

Skin and subcutaneous tissue disorders: angiooedema, bruising, urticaria, pruritus, rash, erythema nodosum.

Musculoskeletal and connective tissue disorders: arthralgia, myalgia including muscle cramps.

General disorders and administration site conditions: asthenia/fatigue, malaise, oedema, pyrexia.

Very rare cases of Churg-Strauss Syndrome (CSS) have been reported during montelukast treatment in asthmatic patients.

Other reported adverse events include:
Cough, wheezing, sinusitis, pharyngitis, laryngitis, rhinitis (infective), acute bronchitis, otitis, otitis media, pneumonia, influenza, rhinorrhea, nasal congestion, ear pain, dental pain, gastroenteritis (infectious), pancreatitis, pyuria, viral infection, and varicella, atopic dermatitis, tooth infection, skin infection, and myopia, eczema, dermatitis, and conjunctivitis somnolence, trauma,

USE IN SPECIAL POPULATIONS

Pregnancy
Montelukast has not been studied in pregnant women. Montelukast should be used during pregnancy only if it is clearly needed.

From worldwide post marketing experience, congenital limb defects have been rarely reported in offspring of women being treated with montelukast during pregnancy. Most of these women were also taking other asthma medications during their pregnancy. A causal relationship between montelukast and malformations (i.e. limb defects) has not been established.

Lactation

It is not known if montelukast is excreted in human milk. Because many drugs are excreted in human milk, caution should be exercised when montelukast is given to a nursing mother.

Elderly

In reported clinical studies, there is no age-related difference in the efficacy or safety of montelukast.

Hepatic Insufficiency

No dosage adjustment is required in patients with mild-to-moderate hepatic insufficiency.

Renal Insufficiency

No dosage adjustment is recommended in patients with renal insufficiency.

OVERDOSAGE

No specific information is available on the treatment of overdose with montelukast.

Treatment may include removal of unabsorbed material from the gastrointestinal tract, clinical monitoring and supportive therapy if required. It is not known if montelukast can be removed by peritoneal dialysis or hemodialysis. Patients in whom intentional overdose is confirmed or suspected should be referred for psychiatric consultation.

STORAGE

Store below 30ºC, protect from light and moisture

KEEP ALL MEDICINES OUT OF REACH OF CHILDREN

PACKING

Blister strips of 3 x 10’s, 6 x 10’s and 10 x 10’s

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Product Registration Holder/ Manufactured by:
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